Process Characterization Plan

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCP-004

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ALCOA+, attributable, legible, contemporaneous, original and accurate data principles; AMV, analytical method validation; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; ELISA, enzyme-linked immunosorbent assay; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; NOR, normal operating range; NTU, nephelometric turbidity unit; PCP, Process Characterization Plan; PCR, Process Characterization Report; qPCR, quantitative polymerase chain reaction; rcf, relative centrifugal field; RSD, relative standard deviation; SEC, size-exclusion chromatography; SOP, standard operating procedure; WC-CPP, well controlled critical process parameter.

1 Purpose and scope

Harvest and clarification is the primary recovery step of the A-Mab drug substance process, in which the antibody is separated from the cells and the cell debris of the production bioreactor and a clarified feed is delivered to Protein A capture. The step forms no product quality attribute. What it does set is the particle load and the impurity burden that the capture step receives, and that input shapes how the rest of the purification train performs.

This plan defines the characterization studies that will support the commercial operating ranges for the step. It covers the parameters in Table 6.1, each studied one at a time on a qualified scale-down model held at commercial equivalent conditions, and it measures product recovery, post-clarification turbidity, depth filter throughput and the impurity burden delivered to capture at every condition. The plan is prospective, so no results appear in it, and the outcome is reported in PCR-004.

Three things sit outside the scope. The production bioreactor and the Protein A capture step are characterized under PCP-003 and PCP-005, and this plan addresses them only at the interface each shares with harvest. No viral clearance is claimed for harvest or for clarification, so the step contributes nothing to the modular clearance strategy set out in PCMP-001 (International Council for Harmonisation 2023a). Confirmation of the step at commercial scale belongs to Stage 2 of the process validation lifecycle (U.S. Food and Drug Administration 2011) and is delivered by process performance qualification.

2 Objectives

The characterization has five objectives, and each is written so that PCR-004 can state plainly whether it was met.

- (i) Quantify the relationship between each parameter and the quality of the clarified harvest, across the characterization ranges given in Table 6.1.
- (ii) Determine a proven acceptable range for each parameter against the acceptance basis in §7.
- (iii) Show that each normal operating range sits inside its proven acceptable range, with margin at every edge where the characterization range extends past it.
- (iv) Provide the evidence needed to classify each parameter under SOP-4001.
- (v) Describe the clarified harvest that the step delivers at each condition studied, so that this study and the capture characterization (PCP-005) can be read together.

The first four objectives follow the pattern used for every step in the campaign (PCMP-001). Objective (iii) carries a bound because the characterization range and the normal operating range of post-clarification turbidity share their lower edge (Table 6.1), and no margin can be shown at an edge where the two ranges meet. The fifth objective is specific to primary recovery. Harvest hands the purification train its feed, so a description of that feed at each condition studied is part of what the step has to deliver, alongside the operating ranges themselves.

3 Related documents

Table 3.1 sets out the documents this plan depends on and the documents it feeds. RA-001 fixed the study scope for Step 4 before any study was designed. PCMP-001 states the campaign approach that this plan instantiates. PCR-004 is the deliverable, and PCMR-001 consolidates it with the reports for the other steps.

Table 3.1: Documents related to this plan.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-004	Process Characterization Report (Harvest and Clarification (Step 4))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2 lists the controlled procedures and the validated analytical methods used in the study. Performance of the methods that carry the quantitative decisions is summarized in Appendix B.

Table 3.2: Controlled procedures and validated analytical methods for Step 4.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit operation description and prior knowledge

Harvest and clarification sits between the production bioreactor and Protein A capture, and it is the only step of the train that handles whole cells (Table 4.1). Everything downstream of it operates on a clarified solution.

Table 4.1: The A-Mab drug substance process train and the principal role of each step.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

The step runs as two unit operations in series. Continuous disk-stack centrifugation removes the cells and the larger debris from the culture broth, depth filtration then removes the fines that the centrifuge leaves behind, and a sterilizing grade filter closes the train and delivers the clarified harvest to the capture feed vessel. Separation in the centrifuge is driven by the density difference between the cells and the medium, and it is governed by the relative centrifugal field (rcf) and by the flow per unit settling area. Depth filtration works by a combination of retention within the media matrix and adsorption of fines onto the media, so its performance falls away as the media approaches its capacity.

The platform recovery step has been used for three prior platform products (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical supply, and recovery across the step is about 95% on the platform, with a coefficient of variation of about 2% between batches. The clarified harvest carries a host cell protein burden of the order of 500,000 ng/mg into capture, which is the load the capture step receives. This body of experience is what allows the step to be bounded by univariate studies instead of a designed experiment.

Three mechanisms frame the study. Raising the centrifugal field improves solids removal and lowers centrate turbidity, but it also raises the hydrodynamic shear the cells experience, and shear releases intracellular protein and DNA into the broth. The two effects run in opposite directions, so the useful range of the field is bounded at both ends and neither end can be assumed from the platform alone. Raising the depth filter load moves the media towards its capacity, at which point fines break through and the differential pressure across the train rises, so the load has to be studied beyond the value the platform uses in order to locate that point. Turbidity is treated as a parameter in its own right, because the clarified harvest is released to capture against a turbidity limit. The study has to describe what the harvest carries at each level of turbidity, which is what such a limit is set against.

4.2 Quality attributes in scope

No drug substance quality attribute is formed at this step, and none is cleared by it. Three attributes are nevertheless in scope, because the clarified harvest carries them into the purification train (Table 4.2).

Table 4.2: Quality attributes carried by the clarified harvest into the purification train.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60

Host cell protein (HCP) is the attribute most exposed to this step. It is of moderate to high criticality, and the burden entering capture is set both by the culture that produced it and by how much intracellular material primary recovery releases. Residual DNA behaves the same way, and its limit is set per dose where the host cell protein limit is a ratio against product (Table 4.2). Its criticality is very low, because the clearance available downstream of this step is large. Aggregate is of high criticality and is formed in the bioreactor (PCR-003), and it is expected to pass through recovery unchanged, which the study will confirm and not assume.

None of the three attributes is controlled by this step alone, and none is controlled by it principally. Host cell protein and residual DNA are cleared by Protein A capture (PCR-005), by cation exchange (PCR-007) and by anion exchange (PCR-008), and aggregate is polished at cation exchange (PCR-007). The contribution of harvest is to bound the burden that enters capture and to describe it at every condition studied. Criticality is treated as a continuum here, in the sense of ICH Q9 and of the A-Mab case study, and the levels in Table 4.2 follow the Tool #1 assessment of impact and uncertainty (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b).

4.3 Risk-based prioritization of parameters

RA-001 ranked every parameter of the step before the studies were designed, and its output is the study scope this plan executes. The ranking combines the potential impact of a parameter on a quality attribute with its potential to interact with other parameters. For every parameter of the step, RA-001 recorded an effect on process performance and on the quality of the feed delivered to capture, and no impact on a drug substance quality attribute. That placed each of them below the band that calls for a multivariate design, and each was assigned to univariate assessment (the parameters that do reach that band at other steps are studied by designed experiment). The reasoning behind the split is set out in RA-001 and in PCMP-001.

Univariate assessment is the right instrument here for a second reason. The parameters act through different mechanisms, and in two cases they act through different equipment. The centrifugal field acts on the separation of cells from broth, and the depth filter load acts on the retention of fines by the media. Turbidity is the outcome of the two operations taken together, and it is controlled by them. An interaction that changed a quality outcome would need two of these mechanisms to act on the same attribute at the same time, and no such pathway has been seen on the platform or in the development work that supports the transfer (PTP-001). §12 states the assumption this rests on and the observation that would falsify it.

The characterization ranges in Table 6.1 are wider than the normal operating ranges, and that is deliberate. A range that covers only routine operation cannot show where performance begins to fail, and it leaves no room for later movement inside a known space (CMC Biotech Working Group 2009). Each range is bounded by a mechanism. The centrifugal field is studied from 6000 to 12000 × g, which spans the field at which centrate turbidity is expected to rise at the low end and the field at which shear damage is expected to become measurable at the high end. The depth filter load is studied from 80 to 160 L/m^2, and its upper edge sits well above the 140 L/m^2 that bounds routine operation, so that the load at which fines break through falls inside the range studied. Turbidity is studied from 1 to 20 NTU, which reaches well beyond the upper edge of the normal operating range. The levels above that edge give measured material at turbidities a release limit would reject, so that an excursion can be assessed against data instead of assumption. What those levels are judged on is set out in §7.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be run on a scale-down model of the recovery train, qualified against commercial scale data under SOP-1001. The model has two parts, because the step has two unit operations, and each part is scaled on the variable that governs its mechanism.

The centrifuge model reproduces the commercial separation on a bench scale device operated at matched equivalent settling area, so that the flow per unit settling area is the same at both scales. Shear is handled separately, and it is handled first. The cells are exposed to the energy dissipation rate of the commercial feed zone in a rotating shear device before they are settled, because the feed zone and not the bowl is where the damaging shear occurs at commercial scale. The depth filtration model uses the same media grade as the commercial train and is scaled at constant area normalized load, with matched flux and with the same flush and recovery volumes per unit area.

Qualification will compare the model against commercial scale data at the set-point condition for product recovery, centrate turbidity, post-clarification turbidity, host cell protein and residual DNA. The model is qualified when the difference between the scales in each of these measures falls inside the intermediate precision of the method that measured it, at the confidence level stated in SOP-1001. The criterion is equivalence within a stated bound and not the absence of a significant difference, because a small qualification run set would pass the second on low power alone. The number of qualification runs, the confidence level and the test are defined in SOP-1001, and the qualification record will be cited in PCR-004.

The warrant the model has to carry is narrow. It supports claims about operating ranges and about the quality of the clarified harvest at those ranges. It does not reproduce the hold times, line volumes or transfer rates of the commercial facility, which are addressed in the transfer plan (PTP-001). It supports no claim about viral clearance, and none is made.

5.2 Operation

The step will be operated as SOP-2005 and SOP-2006 describe it, at commercial equivalent conditions. The feed is the harvested culture broth taken at the end of the fed-batch production culture, under the conditions defined in PCP-003. Broth is fed continuously to the disk-stack centrifuge at the set-point field of 9000 × g. The centrate passes to the depth filter train at the set-point load of 120 L/m², then through a sterilizing grade filter into the capture feed vessel.

Inputs that are controlled to platform specification are not characterized here. The depth filter media grade, the flush and hold buffers and the operating temperature are identity and quality controlled under SOP-2006. Media lot and buffer lot are recorded for every run, so that a lot effect can be separated from a parameter effect if one appears.

5.3 Analytical methods

The study is supported by 4 validated methods, and each measures one response. Turbidity in the centrate and in the clarified pool is measured by nephelometry (AMV-3015). Host cell protein is measured by ELISA (AMV-3012) and residual DNA by qPCR (AMV-3014), on the harvested broth and on the clarified pool, so that the burden delivered to capture is always reported against the burden presented to the step. Aggregate is measured by size-exclusion chromatography (AMV-3011) across the step, to confirm that the size variant profile passes through unchanged.

Performance of the methods that carry the quantitative decisions is given in Appendix B. The turbidity method has an intermediate precision of 4% and a limit of quantitation of 0.5 NTU, and that limit sits below the lower edge of the normal operating range for turbidity, which is 1 NTU, so the whole of the studied range can be measured and not merely bounded. The host cell protein ELISA has an intermediate precision of 9.5% and a limit of quantitation of 1 ng/mg. Of the methods whose validated performance is summarized in Appendix B, it is the less precise, and §7 sets the decision rule that follows from that. Performance of the residual DNA and size-exclusion methods is held in their validation reports, AMV-3014 and AMV-3011.

5.4 Sampling plan

Three samples will be taken from every run. The first is the harvested broth presented to the centrifuge, the second is the centrate leaving it, and the third is the clarified pool after depth and sterilizing filtration. Turbidity is measured on the centrate and on the clarified pool. Host cell protein, residual DNA and aggregate are measured on the harvested broth and on the clarified pool, and volume and protein concentration are recorded at all three points, which is what closes the mass balance and gives step recovery for the run.

Differential pressure and throughput are recorded continuously across the depth filter train for the whole of each run. The shape of the pressure curve is the earliest indication that the media is approaching its capacity, and it is available before any assay result is. Retain samples are held for every run under SOP-2006 and may be re-tested if an analytical result is questioned.

5.5 Statistical methods

The analysis is univariate, because the design is univariate. Each response will be regressed on the parameter of its series across that parameter's characterization range. A linear term is fitted first, and a quadratic term is added only where the residuals show curvature. The slope is tested at $\alpha = 0.05$. Where the slope does not differ significantly from zero, the response is declared insensitive to the parameter over the range studied, and the null result is reported instead of dropped, because a parameter with no effect across a wide range is itself evidence that the step is robust to it.

The set-point runs, one of which sits in each of the 3 series, carry two jobs, and the two are taken in order. They are first compared against each other for a lot effect, by the rule §7 gives, because each series uses its own broth lot. Where no lot effect is found they are pooled into the estimate of variability between runs, which carries 2 degrees of freedom and is the reference against which every parameter effect is judged. Where a lot effect is found the affected series is repeated, and the estimate is formed once the repeat is in hand, since a lot difference left inside it would inflate the yardstick every effect is measured against.

Statistical significance and practical significance are separated explicitly. An effect smaller than the intermediate precision of the method that measured it will be reported as detected but not meaningful, and PCR-004 will say so in those terms. This matters most for host cell protein, where the assay contributes about 40.0% of the total observed variance, and for turbidity, where it contributes about 38.0%.

No process capability index will be calculated for this step. A capability index is a statement about a quality attribute against its acceptance criterion, and this step has no attribute of its own. Commercial scale capability for the attributes the clarified harvest carries is estimated across the whole train from a simulation of 2,000 batches with every parameter inside its normal operating range, and it is reported in PCMR-001.

6 Study design

6.1 Factors, ranges and study type

The design bounds each parameter on its own and claims nothing about interactions between them. Table 6.1 gives the set-point, the normal operating range, the characterization range and the study type for each.

Table 6.1: Parameters to be studied, with set-points, normal operating ranges, characterization ranges and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Centrifugation (rcf)	g	9,000	6000–12000	8000–10000	univariate
Depth filter load	L/m2	120	80–160	100–140	univariate
Post-clarification turbidity	NTU	5	1–20	1–10	univariate

The set-point is the commercial target and the centre of every series, and the normal operating range is the band within which routine manufacture is held. The characterization range is the wider band the study will explore, and the characterization ranges taken together are the knowledge space for the step. The operating region proposed in PCR-004 will be a subset of that space and cannot be larger than it.

6.2 Univariate assessment

No screening design and no response-surface design are planned for this step. At the steps that carry a designed experiment, the screening design identifies which parameters have an effect and the response-surface model is the predictive model and the basis of the design space (PCMP-001). Neither applies here. No quality attribute is formed or cleared at harvest, the parameters are assessed one at a time, and nothing in this plan will support a predictive multivariate model of the step.

The study is a set of 3 univariate series, one per parameter, executed on the qualified scale-down model. Within a series the parameter of interest is set in turn at the two edges of its characterization range, the two edges of its normal operating range and its set-point. The other parameters that can be set are held at their set-points, and post-clarification turbidity, which is an output of the clarification train and not a setting, is measured in the two series that do not challenge it. Where two of those levels coincide the condition is run once. That happens for post-clarification turbidity, whose normal operating range and characterization range share a lower edge, so that series carries one run fewer than the other two. The full schedule of 14 runs is given in Appendix A.

The turbidity series is a challenge series and is executed differently from the other two. Turbidity is an output of the centrifuge and the depth filter train and cannot be set directly, so both operations run at their set-points and the clarified pool is brought to each level by adjusting the clarification train. An additional filtration stage is used to reach the levels below the set-point, and unclarified centrate is blended into the pool to reach the levels above it.

Placing the set-point in every series is deliberate. It gives 3 replicate runs at one condition, spread across the whole execution instead of clustered at the start, so the estimate of variability between runs also detects drift in the feed material or in the model. Run order will be randomized within a series, and each series will use a single harvested broth lot, so that variation in the feed cannot be confounded with the parameter being varied.

A confirmation set follows the univariate series. For each parameter, the proposed edges of its proven acceptable range will be run again with the other parameters held at the corner of their normal operating ranges that the univariate results identify as least favorable (the direction of that corner is fixed from the effects seen in the series). The choice is recorded before the confirmation runs are executed. These runs are the direct test of the assumption that the parameters act independently, and §12 states what will happen if that assumption fails.

7 Acceptance and decision criteria

The criteria below are fixed before execution, and PCR-004 will report against them in the order given here.

The scale-down model is qualified when the comparison described in §5.1 shows the model and commercial scale to agree, within the precision of the method that measured each one, on recovery, turbidity, host cell protein and residual DNA. Studies executed on an unqualified model are not reportable. Qualification must therefore be closed before the first series starts.

A run is valid when every analytical result falls inside the validated range of its method, and when the run was executed within the tolerances given in SOP-2005 and SOP-2006. A series is valid when its set-point run agrees on recovery with the set-point runs of the other series. The platform standard deviation of step recovery is 1.9 recovery points, and each set-point run must sit within three of those standard deviations, which is 5.7 recovery points, of the mean of the set-point runs. The comparison is made against the platform standard deviation because a single set-point run in each series leaves no within-series estimate to test against.

A parameter level is acceptable when all of the following hold. Step recovery is at or above 89.3%, which is the platform mean less three times its standard deviation between batches. Turbidity of the clarified pool is at or below 10 NTU, the upper edge of the normal operating range for that attribute. Host cell protein and residual DNA in the clarified pool are not raised above the set-point condition by more than the intermediate precision of the method that measured them. Aggregate across the step is unchanged within the validated precision of AMV-3011.

Two of these criteria do not apply in the turbidity series, where the pool turbidity is the condition set and not a response of the step. The turbidity criterion is the first of them, since a level of that series would otherwise be judged against itself and the acceptable range could only come back as the normal operating range. The criterion on host cell protein and residual DNA is the second, because the levels above the set-point are reached by blending unclarified centrate into the pool, which raises both by construction. A level of the turbidity series is acceptable when recovery meets its floor and when aggregate across the step is unchanged. The impurity burden measured at each level is reported and not judged, and it is the description of the clarified harvest that objective (v) calls for.

The proven acceptable range of a parameter is the contiguous span of that parameter, containing its set-point, over which every criterion that applies to it is met in the univariate series and again in the confirmation runs. §8 describes how each range will be determined and reported.

Classification follows SOP-4001 and uses the definitions given in the A-Mab case study (CMC Biotech Working Group 2009). A parameter whose variation across its characterization range affects a critical quality attribute is a critical process parameter, and it is a well controlled critical process parameter where the risk of leaving the design space is low. A parameter that affects process performance but no quality attribute is a key process parameter. A parameter that affects neither is a general process parameter. The class of each harvest parameter is an outcome of the study and is stated in PCR-004, not here.

The study fails an objective when a proven acceptable range does not contain its normal operating range with the margin objective (iii) states. In that case the normal operating range will be narrowed to what the data support, or the step will be re-designed and re-characterized, and the decision and its basis will be recorded as a deviation in PCR-004. A proven acceptable range that turns out wider than expected is not a failure, and it does not license operation outside the characterization range that produced it.

8 Proven acceptable ranges (planned analysis)

The acceptance basis has to be stated before the analysis, because this step has no specification of its own. No drug substance specification applies directly to the clarified harvest. For a step that carries a viral clearance attribute the step level requirement would be back-calculated from the cumulative requirement less the clearance credited to the other steps, and no clearance is credited to harvest, so that route does not apply here. The basis is the set of criteria in §7. Those criteria are the recovery floor, the turbidity limit on the clarified pool, the bound on how far host cell protein and residual DNA may rise across the step, and the requirement that aggregate pass through unchanged. Which of them applies to a given series is stated there.

Two analyses will be run for each parameter. The first is the univariate series itself, in which the other parameters that can be set are held at their set-points and the parameter of interest is scanned across its characterization range. The second is the confirmation set described in §6.2, in which the other parameters sit at the least favorable corner of their normal operating ranges. The second analysis is executed and not simulated. At the steps that carry a designed experiment, the equivalent analysis propagates the normal operating ranges through a fitted response-surface model by Monte Carlo simulation, but no such model exists at this step, so the corner is executed as a set of runs instead.

The decision rule is the one given in §7. A proven acceptable range is the contiguous span containing the set-point over which every criterion that applies to the parameter is met under both analyses. Where the two analyses give the same span, the parameter is robust to the others moving anywhere inside their normal operating ranges, and PCR-004 will report it in those terms. Where the confirmation runs narrow the span, the narrower span is the proven acceptable range, and the report will name the criterion that binds it and the margin that remains at the set-point.

Each proven acceptable range will be reported with a plot of the governing response against the parameter, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the normal operating range marked on the parameter axis.

This step has no multivariate design space, because none was studied and none is claimed. The operating region for harvest is the intersection of the proven acceptable ranges, which is a univariate construction and not the multivariate region ICH Q8 describes (International Council for Harmonisation 2009). Every range will be bounded by the characterization range that produced it and by the scale-down model that produced the data. A setting outside a characterization range lies outside the knowledge space and is not covered by this study.

9 Data management and integrity

Study data will be recorded and retained under the ALCOA+ principles, so that every result is attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available. Raw analytical data are captured in the validated laboratory information system, and process data are captured in the electronic batch record for each run.

Calculated results, including step recovery and every regression, are produced by a version controlled analysis script and are traceable to the raw data that fed them. Entries are reviewed by a second person before a data set is closed. Corrections are made so that the original entry stays readable, with the reason and the date recorded against it. Access rights and audit trails are managed under the site quality system (International Council for Harmonisation 2008).

10 Roles and responsibilities

Process Development owns this plan, executes the studies and writes PCR-004. Analytical Development runs the validated methods, releases the results and owns the method performance figures quoted in §5.3. Manufacturing Science reviews the scale-down model and the operating conditions for commercial equivalence, and it owns the interface to the receiving site. Statistics reviews the design and the analysis, including the decision on whether a quadratic term is warranted. Quality Assurance approves this plan and the report.

Any departure from this plan during execution is recorded at the time, assessed for impact by Process Development and Quality Assurance, and reported in PCR-004. A departure that invalidates a run requires that run to be repeated before the series is analysed.

11 Deliverables and schedule

The deliverable of this plan is PCR-004, the characterization report for the step. It will present the executed design, the univariate results, the proven acceptable ranges, the parameter classifications, the contribution of the step to the control strategy and any deviations recorded during execution. PCR-004 rolls up into PCMR-001, which consolidates the campaign.

The work runs in four phases, and no phase starts before the previous one is closed. Scale-down model qualification comes first, and the univariate series follow. The confirmation runs come after the series, because the corner they use is chosen from the series results. Analysis and reporting close the study. No calendar dates are fixed in this plan, and the schedule is held in the campaign plan (PCMP-001) and in the transfer plan (PTP-001).

12 Risks and assumptions

Three risks could affect the conclusions of this study, and each has a defined mitigation.

The first is the representativeness of the scale-down model for shear. The bench device reproduces the energy dissipation rate of the commercial feed zone, but it does not reproduce the residence time distribution of a commercial disk-stack bowl, so the exposure of any given cell to that dissipation rate is not identical at the two scales. Qualification against commercial scale data for host cell protein and residual DNA is the control on this, because both respond to cell lysis. If the model overstates lysis the proven acceptable ranges will be conservative, and if it understates lysis the confirmation at commercial scale in Stage 2 will find it.

The second is variation in the harvested broth between series. Cell density and viability at harvest vary from batch to batch, and both change the solids load the step has to handle. Each series will use a single broth lot, and the set-point run in each series gives a direct comparison between lots. If a set-point run departs from the mean of the set-point runs by more than the tolerance given in §7, the affected series will be repeated.

The third is variation in the depth filter media. Media capacity varies between lots, and the depth filter load series is the series most exposed to it. A single media lot family will be used across the study, and the lot will be recorded for every run. A media lot change during the study is a deviation and will be assessed before results are pooled.

Two assumptions underlie the design. The first is that platform experience with X-Mab, Y-Mab and Z-Mab transfers to A-Mab at this step. The mechanism supports that assumption, since primary recovery separates cells from broth and does not act on the antibody itself. The second assumption is that the parameters act independently across their characterization ranges. The confirmation runs described in §6.2 test it directly. If a confirmation run differs from the univariate result by more than the precision of the method that measured it, the assumption fails, and a multivariate study of the affected pair will be raised through PCMP-001 before any range is proposed.

13 Approvals

This plan takes effect once it has been approved by the functions listed in Table 1. Approval commits those functions to the design, the acceptance criteria and the decision rules it contains. Any change after approval is made by revising this document and not by agreement during execution.

14 Appendices

14.1 Appendix A — Planned univariate condition schedule

Table 14.1 gives the 14 runs of the univariate study, with the setting of each of the 3 parameters in every run. The levels are those defined in §6.2. No results appear here. Turbidity is a set target only in its own series, and in the other two it is a measured response, which the table records instead of showing a set-point that is not set.

Table 14.1: Planned univariate condition schedule (settings only; no results).

Run	Series	Level	Centrifugation (rcf) [g]	Depth filter load [L/m2]	Post-clarification turbidity [NTU]
1	Centrifugation (rcf)	Char. low	6000	120	measured
2	Centrifugation (rcf)	NOR low	8000	120	measured
3	Centrifugation (rcf)	Set-point	9000	120	measured
4	Centrifugation (rcf)	NOR high	10000	120	measured
5	Centrifugation (rcf)	Char. high	12000	120	measured
6	Depth filter load	Char. low	9000	80	measured
7	Depth filter load	NOR low	9000	100	measured
8	Depth filter load	Set-point	9000	120	measured
9	Depth filter load	NOR high	9000	140	measured
10	Depth filter load	Char. high	9000	160	measured
11	Post-clarification turbidity	Char. low / NOR low	9000	120	1
12	Post-clarification turbidity	Set-point	9000	120	5

Run	Series	Level	Centrifugation (rcf) [g]	Depth filter load [L/m2]	Post-clarification turbidity [NTU]
13	Post-clarification turbidity	NOR high	9000	120	10
14	Post-clarification turbidity	Char. high	9000	120	20

14.2 Appendix B — Analytical methods summary

Table 14.2 summarizes the performance of the validated methods that carry the quantitative decisions in this study, which are 2 of the 4 listed in Table 3.2. Performance for the remaining methods is held in their own validation reports.

Table 14.2: Performance of the validated methods that carry the study's quantitative decisions.

Method	Title	Precision (% RSD)	LOQ	LOQ unit	Accuracy (%)
AMV-3015	Turbidity by Nephelometry (NTU)	4	0.5	NTU	98
AMV-3012	Host-Cell Protein by ELISA	9.5	1	ng/mg	95

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